Alteryx Workflow Document

Challenge_447_solution_file.yxmd

Report Create Date : 2026-04-25

Workflow Basic Information

FileName : Challenge_447_solution_file.yxmd
Workflow name : Challenge_447_solution_file
yxmd version : 2024.1
RunE2 : T
Workflow description :

Author Information

Author :
Company :
Copyright :

Workflow Meta information

Setting Name	Value
Root Tool Name	
Tool Version	
Tool In DB	
Tool In DB value	False
Category Name	
Search Tags	
Description Link actual	
Description Link displayed	
Workflow ID	70dcbc91-245d-4c6e-a322-baf27357e019
Platform Workflow ID	
Example Description	
Example File	
Telemetry Previous Workflow ID	8cf3036d-df86-4e27-a1f9-0d0af6065e60
Telemetry Origin Workflow ID	bd4bce6a-58b0-4895-819d-8a9250f426d3

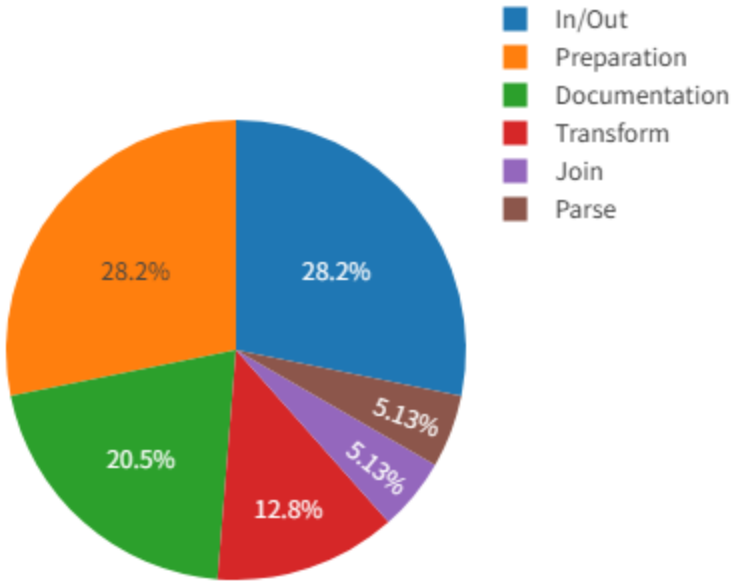
workflow detail setting

Setting Name	Value
yxmdVer	2024.1
RunE2(AMP Engine)	T
Memory_default	True
Global Record Limit	0
Temp Files	True
Annotation on	True
Annotation include Tool Name	False
Convert Error Limit	10
Convert Error Limit Stop	False
Cancel On Error	False
Disable Browse tool	False
Enable Performance Profiling	False
Run With E2	True
Sorted Grouping(Engine Compatibility Mode)	True
Predictive Tools Code Page	1252
Disable All Output	False
Show All Macro Messages	False
Show Connection Status Is On	False
Show Connection Status Only When Runninng	False
Zoom Level	-3
Layout Type	Horizontal
Is Template	False
Name Is File Name	True

Tool statistics

Total Tool Count : 39
Number of tool types : 17

Category



Tool Category	Count
In/Out	11
Preparation	11
Documentation	8
Transform	5
Join	2
Parse	2

Tools

Tool Category	Tool Name	Count
Documentation	Comment	7
In/Out	Browse	6
In/Out	Text Input	4
Transform	Summarize	4
Preparation	Formula	3
Parse	Text To Columns	2
Preparation	Filter	2
Preparation	Sort	2
Documentation	Tool Container	1
In/Out	Input Data	1
Join	Append Fields	1
Join	Find Replace	1
Preparation	Multi-Field Formula	1
Preparation	Multi-Row Formula	1
Preparation	Sample	1
Preparation	Select	1

Tool Category	Tool Name	Count
Transform	Cross Tab	1

Input/Output

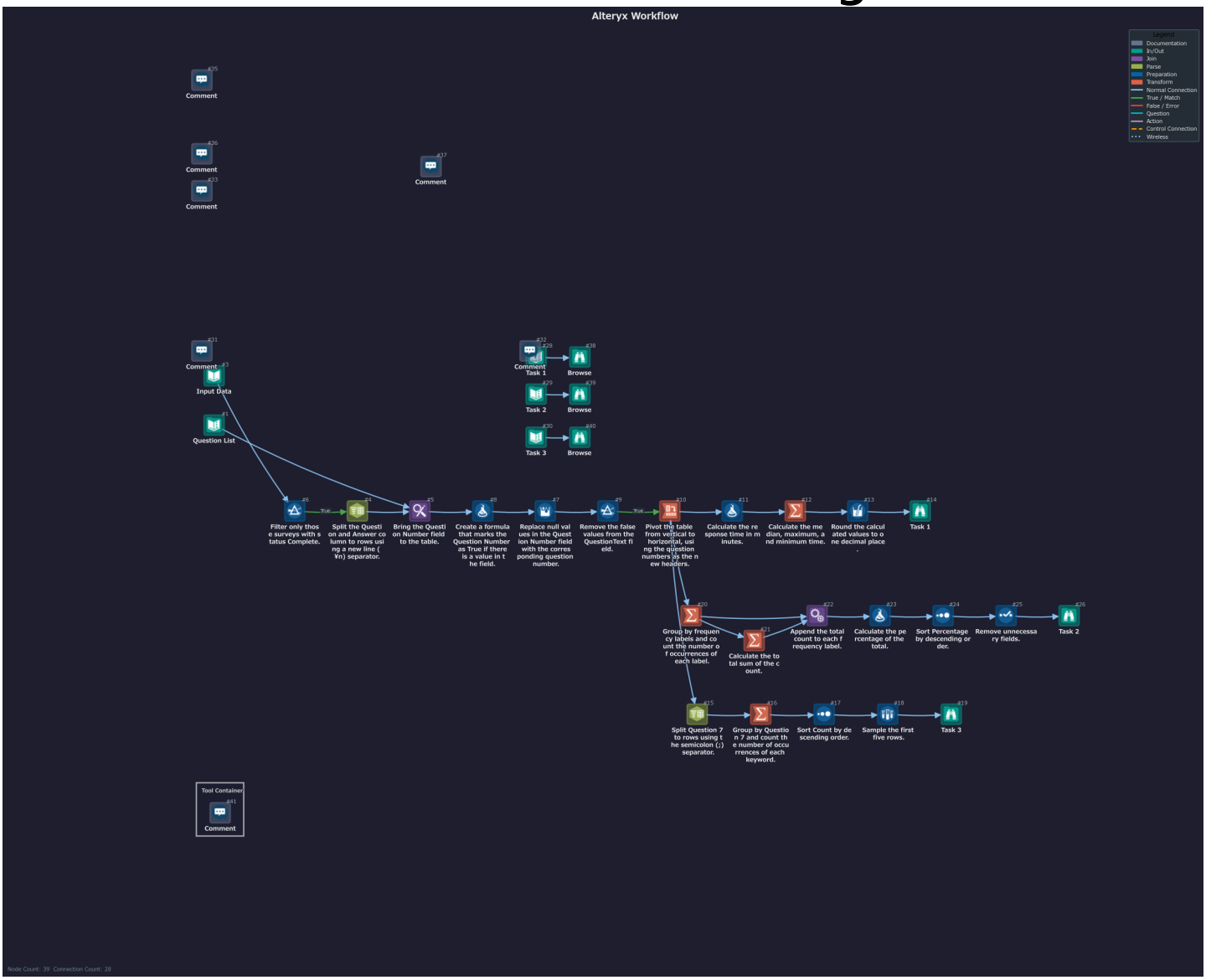
Input overview

Tool Name	ToolID	Input Type	Input
Text Input	1	Text	Text
Input Data	3	Microsoft Excel (*.xlsx)|*.xlsx	questionnaire.xlsx|||`Sheet1$`
Text Input	28	Text	Text
Text Input	29	Text	Text
Text Input	30	Text	Text

Output overview

Tool Name	ToolID	Output Type	Output
Browse	14	Browse	
Browse	19	Browse	
Browse	26	Browse	
Browse	38	Browse	
Browse	39	Browse	
Browse	40	Browse	

Workflow Canvas Image



Alteryx Workflow detail description

Workflow Overview

This Alteryx workflow is designed to analyze survey results regarding Jupyter Notebook usage. The primary objectives are to process raw survey data, which includes handling incomplete entries and merged questions/answers, and then to generate three specific reports. The first report calculates the median, maximum, and minimum time respondents took to complete the survey. The second report determines the count and percentage of respondents based on their Jupyter Notebook usage frequency. The third report identifies the top five words used by respondents to describe Jupyter Notebook. The workflow starts by reading an Excel file containing survey responses and a separate text input with question details, then transforms and aggregates this data to produce the required analytical outputs.

Input/Output Data Overview

Input Data

The workflow utilizes two primary input sources:

Input Source	Description	Key Fields
questionnaire.xlsx (Input Data Tool)	An Excel file containing raw survey responses, where questions and answers are merged into a single field, and survey completion status is recorded.	RecordID, Time Started, Date Submitted, Status, Question_and_Answer
Question List (Text Input Tool)	A manually entered list of question numbers and their corresponding full question text, used for lookup purposes.	Question_No, Question

Output Data

The workflow generates three distinct outputs, each addressing a specific analytical task:

Output (Browse Tool)	Description	Key Fields
Task 1 (Browse 14)	Shows the median, minimum, and maximum survey completion times in minutes, rounded to one decimal place.	Median_Minutes, Min_Minutes, Max_Minutes
Task 2 (Browse 26)	Presents the frequency, count, and percentage of respondents for each Jupyter Notebook usage category (Q1).	Frequency, Count, Percentage
Task 3 (Browse 19)	Displays the top five words used to describe Jupyter Notebook (from Q7) along with their counts.	BestDescribe, Count

Workflow Details

Section 1: Data Preparation and Restructuring

This initial section focuses on ingesting the raw survey data, filtering for complete responses, and transforming the data from a semi-structured format (merged questions and answers) into a clean, tabulated format where each question's answer is in its own column.

Tool Name	Tool ID	Processing Details
Input Data	3	Reads the 'questionnaire.xlsx' Excel file, specifically 'Sheet1$', to bring in the raw survey response data.
Text Input	1	Manually defines a list of question numbers (e.g., Q_1, Q_2) and their corresponding full question text. This data is used as a lookup later.
Filter	6	Filters the incoming survey data to retain only those records where the 'Status' field indicates 'Complete', ensuring only finished surveys are processed.
Text To Columns	4	Takes the 'Question_and_Answer' field from the filtered survey data and splits its content into separate rows wherever a newline character (¥n) is encountered. This effectively unpivots the merged questions and answers into individual records.
Find Replace	5	Performs a lookup operation. It searches the 'Question_and_Answer' field (now containing individual question/answer lines) for matching question text provided by the 'Question List' Text Input. When a match is found, it appends the corresponding 'Question_No' to the current record. This links question numbers to their respective question lines in the unpivoted data.
Formula	8	Creates a new boolean field named 'QuestionText'. This field is set to 'True' if the 'Question_No' field is not null (meaning a question number was successfully found and assigned), and 'False' otherwise.
Multi-Row Formula	7	Populates null values in the 'Question_No' field by looking at the value from the previous row. This is crucial for assigning the correct question number to answer lines that immediately follow a question line. If 'Question_No' is null, it inherits the value from the row above.
Filter	9	Filters the data based on the 'QuestionText' field. Only rows where 'QuestionText' is 'True' (i.e., rows corresponding to actual question statements, now associated with their respective question numbers) are passed through the True output anchor. This removes extraneous rows that are not clearly questions or answers.
Cross Tab	10	Reshapes the data from a long (row-wise) format to a wide (column-wise) format. It groups the data by 'RecordID', 'Time Started', 'Date Submitted', and 'Status'. The 'Question_No' field is used to create new column headers, and the corresponding 'Question_and_Answer' values are populated under these new headers. The aggregation method used for the 'Question_and_Answer' field is 'Concat' with a comma separator, which is suitable for potentially multiple answers or text responses, though in this case it likely results in a single answer per question column.

Section 2: Task 1 - Survey Completion Time Analysis

This section calculates and rounds the median, minimum, and maximum time taken by respondents to complete the survey.

Tool Name	Tool ID	Processing Details
Formula	11	Calculates the difference in seconds between 'Date Submitted' and 'Time Started' fields, then divides by 60 to convert this duration into minutes, storing the result in a new 'Minutes' field.
Summarize	12	Aggregates the data to calculate the median, minimum, and maximum values of the 'Minutes' field, providing key statistics on survey completion time.
Multi-Field Formula	13	Applies a rounding function to the 'Median_Minutes', 'Min_Minutes', and 'Max_Minutes' fields, rounding each value to one decimal place.
Browse	14	Displays the final output for Task 1, showing the rounded median, minimum, and maximum survey completion times.

Section 3: Task 2 - Jupyter Notebook Usage Frequency Analysis

This section analyzes responses to Question 1 ('How often do you use Jupyter Notebook?') to determine the count and percentage for each frequency category.

Tool Name	Tool ID	Processing Details
Summarize	20	Groups the data by the 'Q_1' field (Jupyter Notebook usage frequency) and counts the number of records for each distinct frequency label, creating 'Frequency' and 'Count' fields.
Summarize	21	Calculates the total sum of all 'Count' values generated by the previous Summarize tool, which represents the total number of survey respondents for this question.
Append Fields	22	Attaches the 'Sum_Count' (total respondent count) to every row output from the first Summarize tool, making the total available for percentage calculations.

Tool Name	Tool ID	Processing Details
Formula	23	Calculates a 'Percentage' field by dividing each 'Count' by the 'Sum_Count' and multiplying by 100, then rounding to one decimal place. It also cleans up the 'Frequency' field by setting empty strings to null.
Sort	24	Sorts the results in descending order based on the 'Percentage' field, arranging the usage frequencies from most to least common.
Select	25	Removes the 'Sum_Count' field, which is no longer needed after percentage calculation, to streamline the output.
Browse	26	Displays the final output for Task 2, showing each frequency category, its count, and its percentage of the total respondents.

Section 4: Task 3 - Top 5 Descriptors for Jupyter Notebook

This section processes responses to Question 7 ('Select all words that best describe Jupyter Notebook') to identify the five most frequently used descriptive words.

Tool Name	Tool ID	Processing Details
Text To Columns	15	Splits the 'Q_7' field into separate rows based on the semicolon (;) delimiter. This creates a new record for each descriptive word listed by a respondent.
Summarize	16	Groups the data by the newly split 'Q_7' values (now renamed 'BestDescribe') and counts the occurrences of each unique word, providing a tally for each descriptor.
Sort	17	Sorts the aggregated data in descending order based on the 'Count' field, placing the most frequently mentioned words at the top.
Sample	18	Selects only the first five rows from the sorted data, effectively isolating the top five most common descriptive words.
Browse	19	Displays the final output for Task 3, showing the top five descriptive words and their respective counts.